

Python List - Complete Notes

A beginner-friendly guide to all important List concepts in Python.

1. What is a List?

A list is an ordered, mutable collection that can store multiple values of different data types.

Example:

```
numbers = [10, 20, 30, 40]
```

2. Creating Lists

Empty list: []

With values: [1,2,3]

Mixed: [1, 'Sam', 3.14]

3. Indexing

Positive indexing starts from 0. Negative indexing starts from -1.

4. Slicing

list[start:end:step]

5. Traversing a List

Using for loop and while loop.

6. Important Methods

append(x) - Add one item at end

extend(iterable) - Add multiple items

insert(i,x) - Insert at index

remove(x) - Remove first occurrence

pop() - Remove by index

clear() - Remove all elements

index(x) - Find position

count(x) - Count frequency

sort() - Sort list

reverse() - Reverse list

copy() - Create shallow copy

7. append() Internal Working

Python stores list items in dynamic memory. append() usually adds an item at the end in O(1) average time.

8. extend() Internal Working

extend() iterates through another iterable and appends each element one by one.

9. insert() Internal Working

Elements after the insertion position are shifted right. Time complexity O(n).

10. Common Operations

len(list)
max(list)
min(list)
sum(list)
item in list

11. List Comprehension

squares = [x*x for x in range(10)]

12. Nested Lists

Lists inside lists. Example: matrix = [[1,2],[3,4]]

13. Copying Lists

copy(), slicing [:], and list() constructor.

14. Time Complexities

append() -> O(1) average
extend() -> O(k)
insert() -> O(n)
remove() -> O(n)
search (in) -> O(n)
index() -> O(n)
sort() -> O(n log n)

15. Interview Questions

Find maximum element without max()
Remove duplicates
Find frequency of elements
Reverse a list
Merge two lists
Find second largest element

Summary

Lists are one of the most important Python data structures. Master indexing, slicing, traversal, methods, list comprehensions, and time complexities.